METHODOLOGY PASSPORT CARD

Research Methodology & Experimental Design

1. PROJECT INFORMATION

Research Title: Classification of Textual AI-Phishing Methods in Personal Communications of Private Users

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Repository URL: https://github.com/yeraaaaaah/ai-phishing-classification-benchmark

2. RESEARCH TYPE & JUSTIFICATION

Research Design: Quantitative Multi-Phase Research

Methodological Justification:

- **Phase 1 (Linguistic & Stylistic Analysis):** Quantitative automated extraction of textual properties, including error rate calculations per 100 words, readability indices, and formal vs. emotional tone proportions.
- **Phase 2 (Empirical User Study):** Controlled quantitative experiment measuring human participant detection accuracy (%), trust ratings on a 1–7 Likert scale, and action execution rates (%).

3. FORMULATION OF HYPOTHESES

Hypothesis Set 1: Linguistic & Stylistic Characteristics

- **Null Hypothesis (H_0):** Differences in linguistic coherence, syntactical error rates, and formal tone proportions between LLM-generated phishing texts and traditional manual phishing texts are statistically non-significant.
- **Alternative Hypothesis (H_1):**
  - **H (Coherence):** AI-generated phishing messages exhibit statistically higher linguistic coherence compared to traditional manual phishing messages.
  - **H (Error Rate):** AI-generated phishing messages contain statistically fewer syntactical errors per 100 words compared to traditional manual phishing messages.
  - **H (Style):** AI-generated phishing messages exhibit a statistically higher proportion of formal business/domain-specific style compared to traditional manual phishing messages.

Hypothesis Set 2: Psychological Impact & User Vigilance

- **Null Hypothesis (H_0):** Text generation method and the presence or absence of linguistic anomalies ("red flags") have no statistically significant impact on user spontaneous vigilance, perceived authority, or action execution rates.
- **Alternative Hypothesis (H_1):**
  - **H (Vigilance):** Users' spontaneous vigilance (detection accuracy) when evaluating AI-generated phishing messages is statistically lower than when evaluating traditional manual phishing messages.
  - **H (Authority):** The complete absence of linguistic anomalies statistically increases perceived sender authority ratings.
  - **H (Execution):** Higher perceived sender authority driven by anomaly-free text statistically increases user action execution rates (link clicks or credential entry).

4. VARIABLE MATRIX

Variable Category	Variable Name	Description & Parameters
Independent Variables (IV)	**Generation Method** Presence of Red Flags	LLM-generated (ChatGPT, Claude, Gemini) vs. Traditional Manual Writing. Presence vs. complete absence of linguistic and stylistic anomalies.
Dependent Variables (DV)	**Text Quality Metrics** Stylistic Pattern Ratio Spontaneous Vigilance Action Execution Rate	Error count per 100 words, Readability Index. Proportion of formal business style vs. emotional threat markers. Detection accuracy (%) and trust rating (1–7 Likert scale). Percentage of users attempting link clicks or credential entry.
Controlled Variables (CV)	**Experimental Constants**	Message length and topic domain, target language (English), communication platform/channel.
Covariates	**Participant Demographics**	Participant digital literacy level and age group (measured for statistical control of baseline variance).

5. METRICS & BASELINE SPECIFICATION

Primary Metric:

- **Empirical Primary Metric:** Human Detection Accuracy Percentage (%) — primary operational measure of user vulnerability.

Secondary Metrics:

- **Linguistic Metrics:** Syntactical Error Rate per 100 words & Flesch-Kincaid Readability Index.
- **Behavioral Metrics:** Perceived Sender Authority Rating (1–7 Likert scale) & Action Execution Rate (%).

Guardrail Metrics:

- **Execution Latency:** Benchmark execution time < 2000 ms per text sample.
- **False Positive Rate:** Classification error rate for legitimate messages < 5%.
- **Stability:** 0% unhandled exceptions during pipeline execution.

Baseline & Control Benchmark:

- **Control Corpus:** A curated set of traditional human-crafted phishing emails and messages gathered from open-source threat intelligence (OSINT) databases.
- **Standard Detection Baseline:** Traditional rule-based and keyword-matching detection mechanisms.